# Breast Cancer Project

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## Data Preparation

```r
cancer_raw <- read_csv("Downloads/data.csv", show_col_types = FALSE)

cancer <- cancer_raw %>%
  janitor::clean_names() %>%
  select(diagnosis:fractal_dimension_worst) %>%
  mutate(
    diagnosis = factor(diagnosis,
                       levels = c("B", "M"),
                       labels = c("Benign", "Malignant"))
  )

head(cancer, 5)
```

```
## # A tibble: 5 x 31
##   diagnosis radius_mean texture_mean perimeter_mean area_mean smoothness_mean
##   <fct>           <dbl>        <dbl>          <dbl>     <dbl>           <dbl>
## 1 Malignant        18.0         10.4          123.      1001           0.118
## 2 Malignant        20.6         17.8          133.      1326           0.0847
## 3 Malignant        19.7         21.2          130       1203           0.110
## 4 Malignant        11.4         20.4           77.6      386.          0.142
## 5 Malignant        20.3         14.3          135.      1297           0.100
## # i 25 more variables: compactness_mean <dbl>, concavity_mean <dbl>,
## #   concave_points_mean <dbl>, symmetry_mean <dbl>,
## #   fractal_dimension_mean <dbl>, radius_se <dbl>, texture_se <dbl>,
## #   perimeter_se <dbl>, area_se <dbl>, smoothness_se <dbl>,
## #   compactness_se <dbl>, concavity_se <dbl>, concave_points_se <dbl>,
## #   symmetry_se <dbl>, fractal_dimension_se <dbl>, radius_worst <dbl>,
## #   texture_worst <dbl>, perimeter_worst <dbl>, area_worst <dbl>, ...
```

```r
table(cancer$diagnosis)
```

```
##
##    Benign Malignant
##       356       212
```

The dataset used in this analysis contains measurements of tumor characteristics derived from digitized images of breast masses. The objective is to use these features to classify tumors as either benign or malignant.

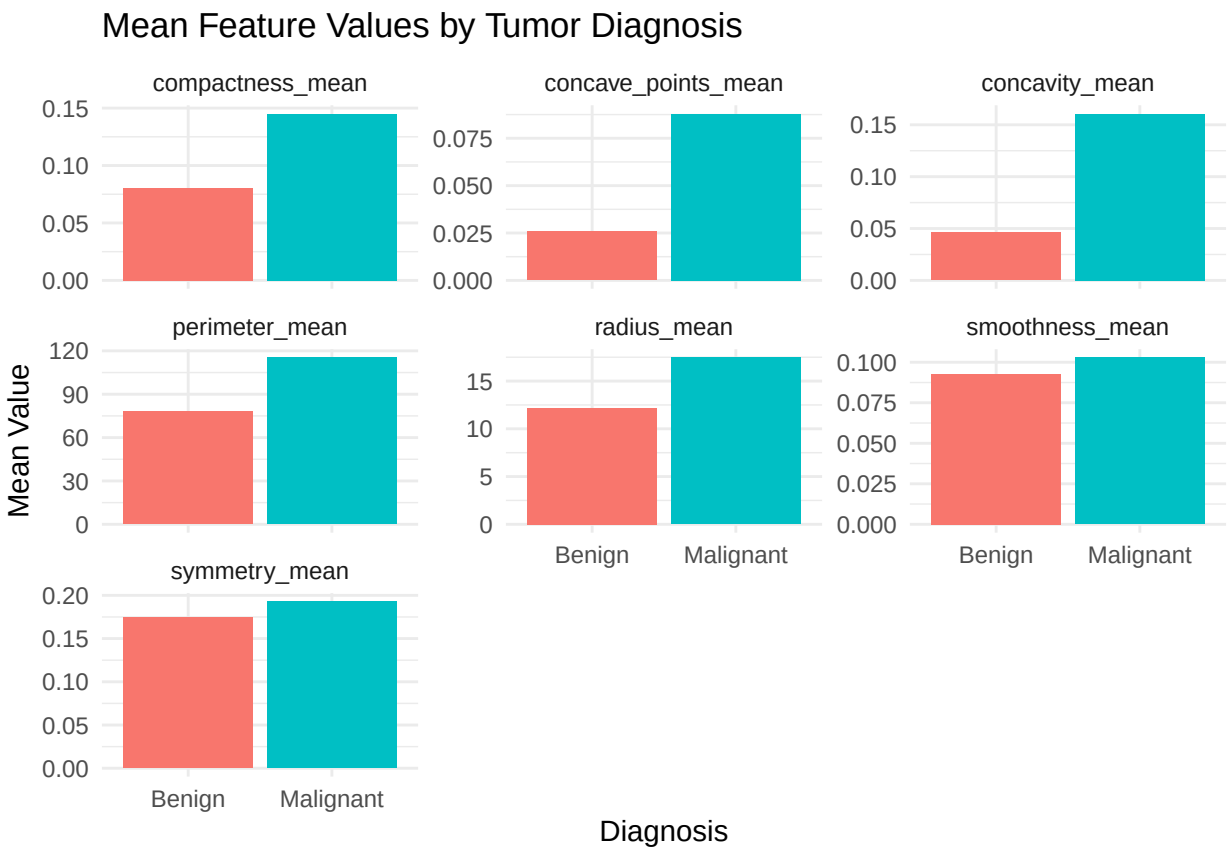
The data was first cleaned and prepared by standardizing column names and converting the diagnosis variable into a factor with clearly defined classes. This ensures consistency in modeling and interpretation throughout the analysis.

## Exploratory Data Analysis

```r
eda_means <- cancer %>%
  group_by(diagnosis) %>%
  summarize(
    radius_mean = mean(radius_mean, na.rm = TRUE),
    perimeter_mean = mean(perimeter_mean, na.rm = TRUE),
    smoothness_mean = mean(smoothness_mean, na.rm = TRUE),
    compactness_mean = mean(compactness_mean, na.rm = TRUE),
    concavity_mean = mean(concavity_mean, na.rm = TRUE),
    concave_points_mean = mean(concave_points_mean, na.rm = TRUE),
    symmetry_mean = mean(symmetry_mean, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(-diagnosis, names_to = "variable", values_to = "mean_value")

# Faceted bar plot of mean values
mean_plot <- ggplot(eda_means, aes(x = diagnosis, y = mean_value,
fill = diagnosis)) +
  geom_col(position = "dodge") +
  facet_wrap(~ variable, scales = "free_y") +
  labs(
    title = "Mean Feature Values by Tumor Diagnosis",
    x = "Diagnosis",
    y = "Mean Value"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

print(mean_plot)
```



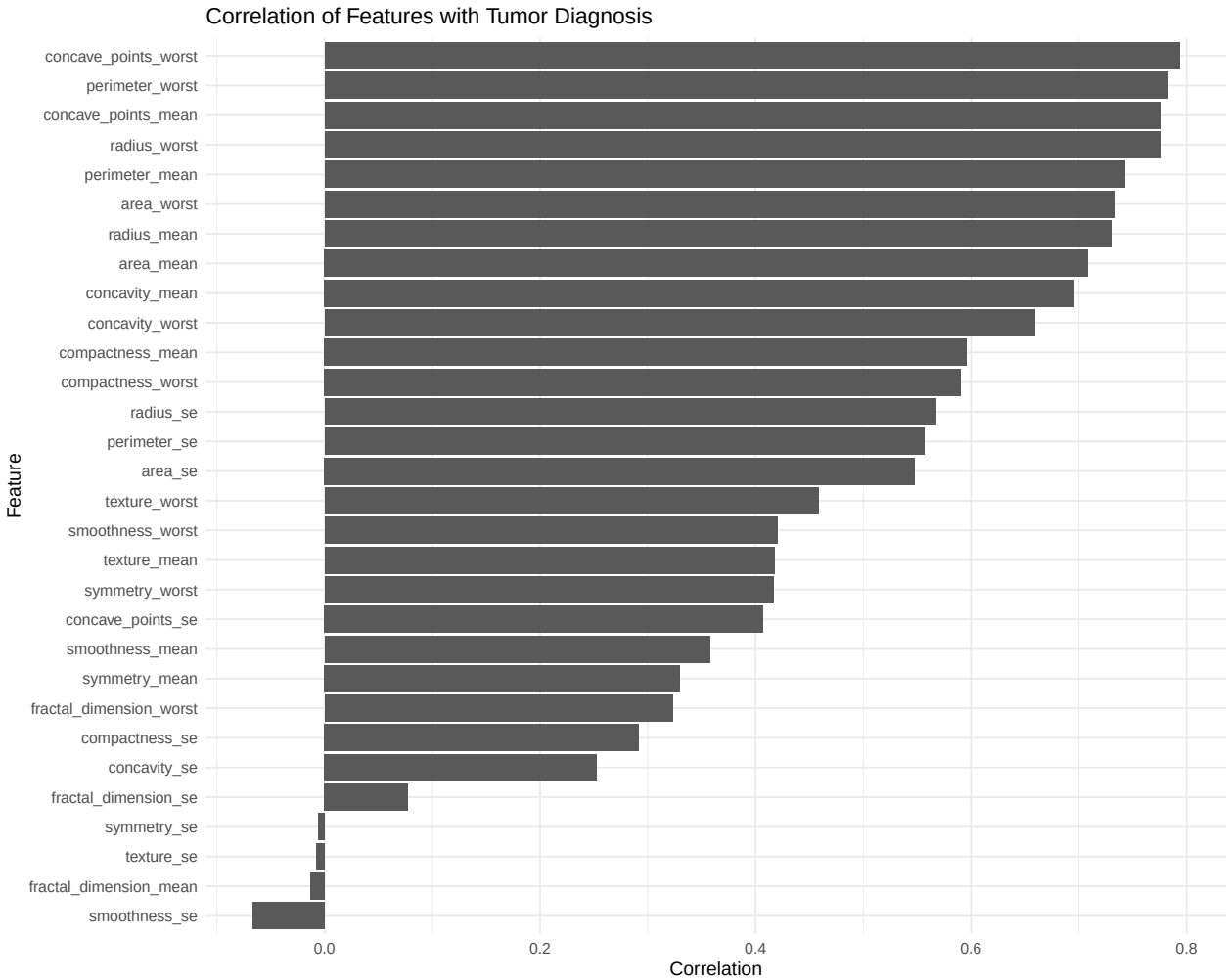

Initial exploratory analysis showed that malignant tumors tended to have higher average values across several features, including radius, perimeter, compactness, concavity, concave points, and symmetry. These differences suggested that the predictor variables may carry meaningful information for distinguishing benign from malignant tumors.

```
numeric_data <- cancer %>%
  mutate(diagnosis = ifelse(diagnosis == "Malignant", 1, 0))

cor_target <- cor(numeric_data, use = "complete.obs")[, "diagnosis"]

cor_df <- data.frame(
  variable = names(cor_target),
  correlation = cor_target
) %>%
  filter(variable != "diagnosis") %>%
  arrange(desc(abs(correlation)))

ggplot(cor_df, aes(x = reorder(variable, correlation), y = correlation)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Correlation of Features with Tumor Diagnosis",
    x = "Feature",
    y = "Correlation"
  ) +
  theme_minimal()
```




Correlation analysis further showed that many size-related variables, including radius, perimeter, and area, were highly correlated with one another. This indicates a significant degree of redundancy within the feature set.

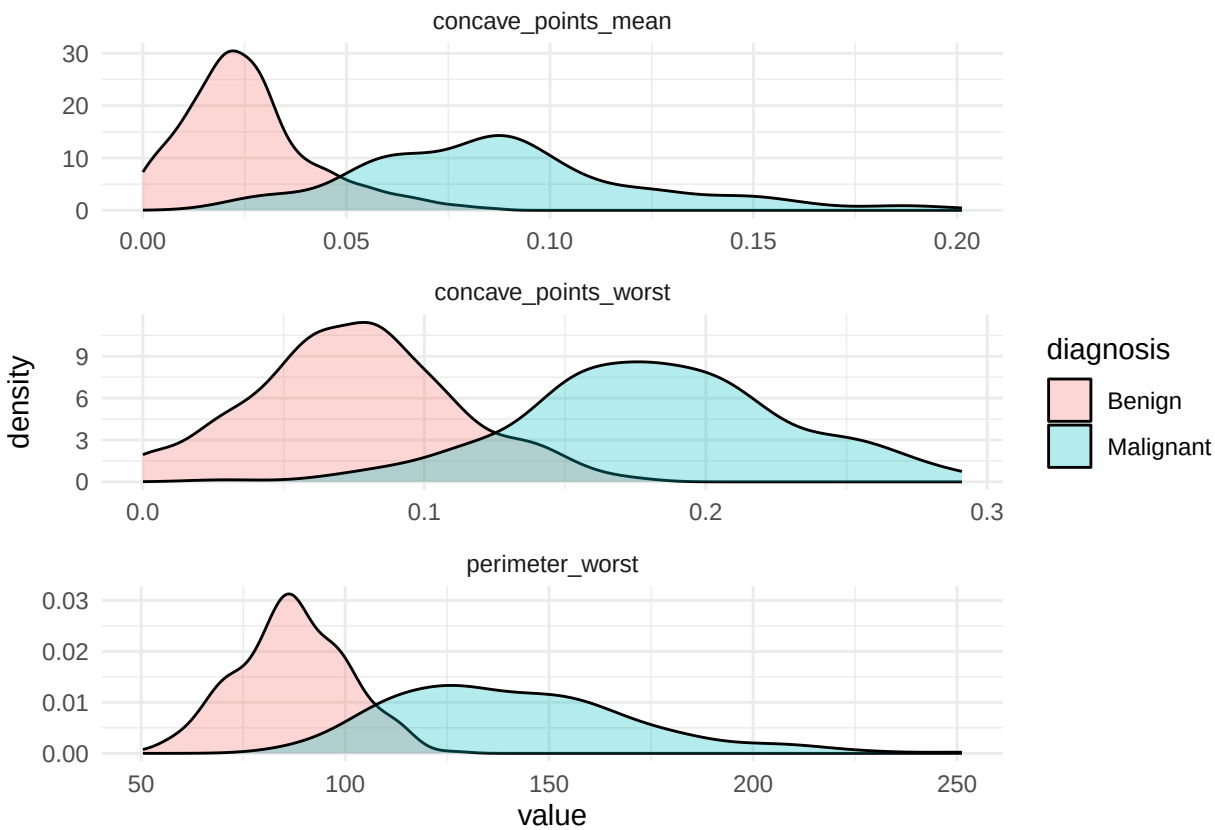
In contrast, features related to tumor shape and boundary irregularity, such as concavity and concave points, also demonstrated strong relationships with diagnosis while being less redundant.

These findings suggest that a smaller subset of variables may be sufficient for accurate classification, and that feature selection could improve model stability.

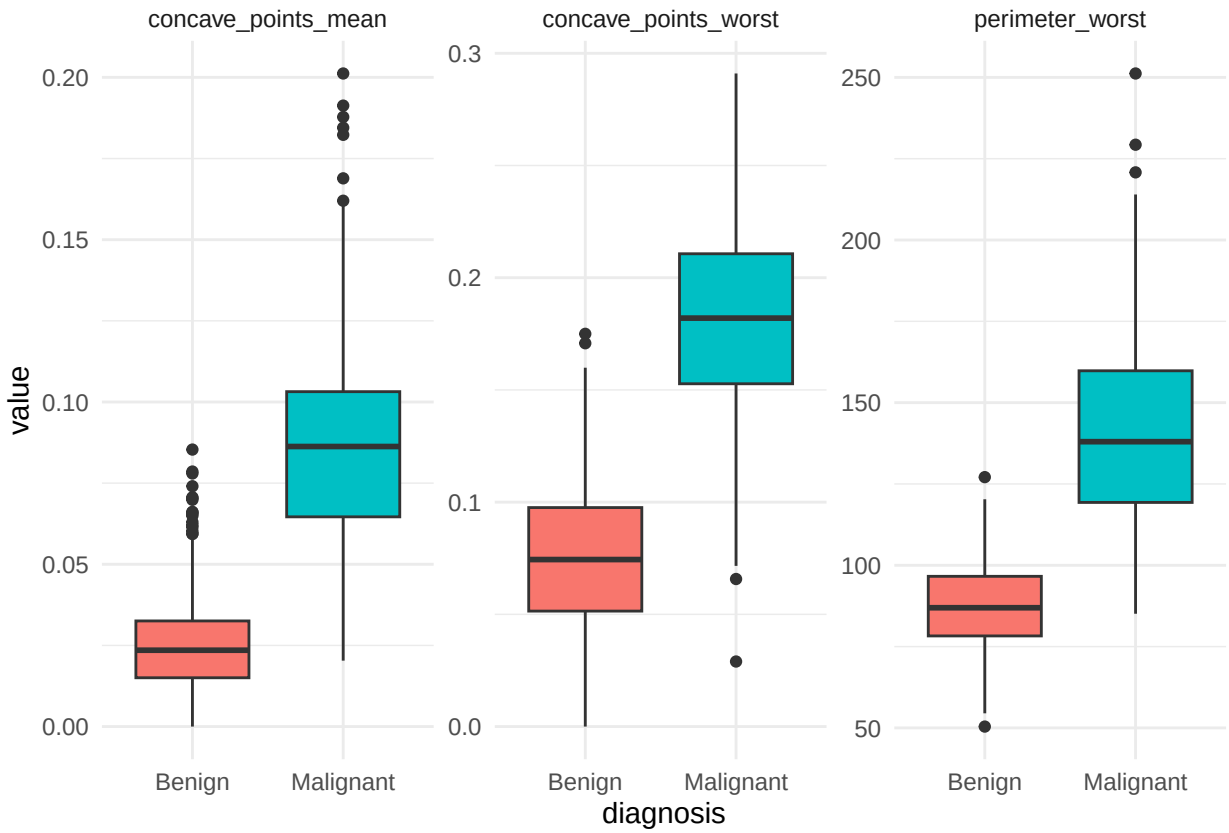
```
top_vars <- cor_df %>%
slice_max(abs(correlation), n = 3) %>%
pull(variable)

plot_data <- cancer %>%
    pivot_longer(
        cols = all_of(top_vars),
        names_to = "variable",
        values_to = "value"
    )
ggplot(plot_data, aes(x = value, fill = diagnosis)) +
    geom_density(alpha = 0.3) +
    facet_wrap(~ variable, scales = "free", ncol = 1) +
```

```
    theme_minimal()
```




```
ggplot(plot_data, aes(x = diagnosis, y = value, fill = diagnosis)) +
    geom_boxplot() +
    facet_wrap(~ variable, scales = "free") +
    theme_minimal() +
    theme(legend.position = "none")
```




Distribution plots of the most strongly correlated variables showed clear separation between benign and malignant tumors. In particular, features such as concave points and radius displayed distinct distributions across the two classes, reinforcing their importance as predictive variables.

Overall, the exploratory analysis indicated that the dataset contains strong predictive signals, but also highlighted the presence of highly correlated features. These observations informed the modeling approach, including the use of regularization techniques to reduce redundancy and improve model interpretability.

## Logistic Regression Model

```
set.seed(123)



train_index <- caret::createDataPartition(cancer$diagnosis, p = 0.8,
list = FALSE)
train <- cancer[train_index, ]
test  <- cancer[-train_index, ]

model <- glm(diagnosis ~ ., data = train, family = binomial)


probs <- predict(model, newdata = test, type = "response")


preds <- ifelse(probs > 0.5, "Malignant", "Benign")
```

```r
preds <- factor(preds, levels = levels(test$diagnosis))

cat("=== Logistic Regression Results ===\n")
```

```
## === Logistic Regression Results ===
```

```r
caret::confusionMatrix(preds, test$diagnosis, positive = "Malignant")
```

```
## Confusion Matrix and Statistics
##
##            Reference
## Prediction  Benign Malignant
##   Benign        64         0
##   Malignant      7        42
##
##                Accuracy : 0.9381
##                  95% CI : (0.8765, 0.9747)
##     No Information Rate : 0.6283
##     P-Value [Acc > NIR] : 1.718e-14
##
##                   Kappa : 0.8717
##
##  Mcnemar's Test P-Value : 0.02334
##
##             Sensitivity : 1.0000
##             Specificity : 0.9014
##          Pos Pred Value : 0.8571
##          Neg Pred Value : 1.0000
##              Prevalence : 0.3717
##          Detection Rate : 0.3717
##    Detection Prevalence : 0.4336
##       Balanced Accuracy : 0.9507
##
##        'Positive' Class : Malignant
##
```

```r
# ROC and AUC
roc_obj <- pROC::roc(test$diagnosis, probs)
```

```
## Setting levels: control = Benign, case = Malignant
```

```
## Setting direction: controls < cases
```

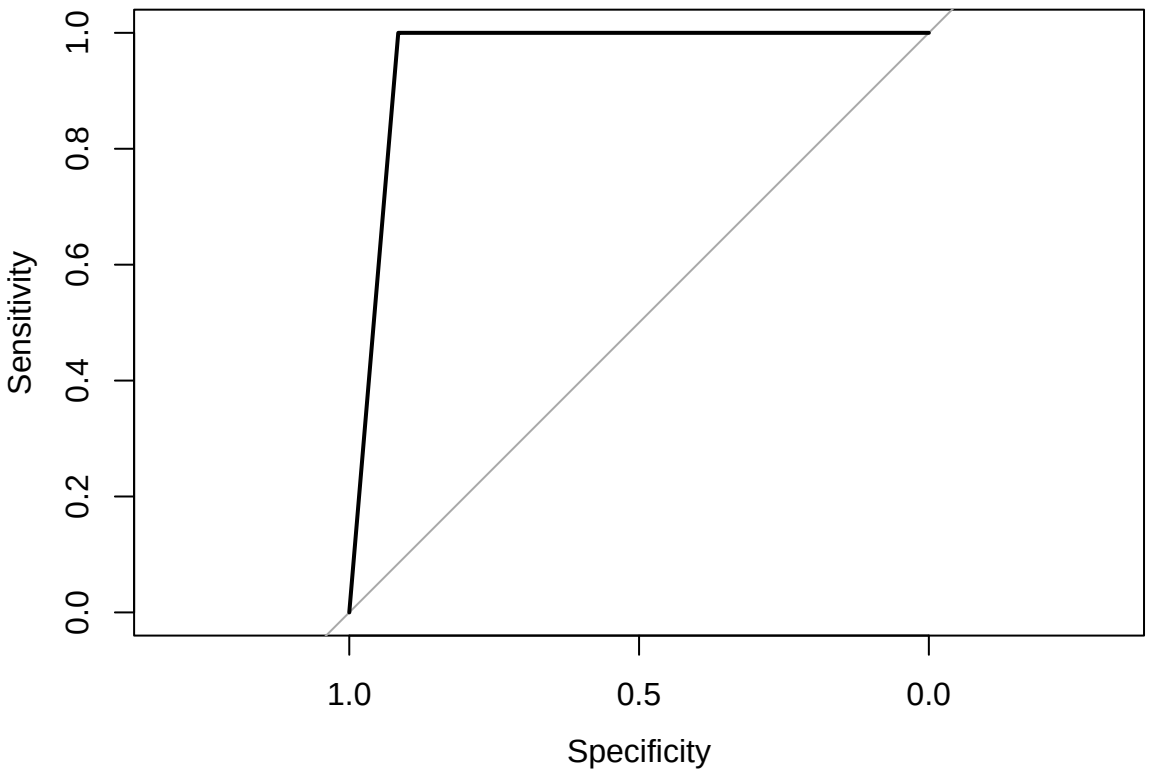
```r
cat("\n=== ROC AUC ===\n")
```

```
##
## === ROC AUC ===
```

```r
pROC::auc(roc_obj)
```

```
## Area under the curve: 0.9577
```

```r
plot(roc_obj)
```




A logistic regression model was used as a baseline approach to classify tumors as benign or malignant. The model was trained on 80% of the dataset and evaluated on the remaining 20%.

The model achieved strong performance on the test set, with high accuracy, sensitivity, and specificity, along with an AUC of approximately 0.96. This indicates that the predictor variables contain substantial information for distinguishing between benign and malignant tumors.

While the logistic regression model performed well, the presence of highly correlated predictors suggested potential redundancy in the feature set. This motivated the use of a regularized modeling approach to improve interpretability and reduce the number of predictors.

## Model Validation

```r
set.seed(123)

control <- caret::trainControl(
  method = "cv",
  number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary,
  savePredictions = TRUE
)

cv_model <- caret::train(
  diagnosis ~ .,
  data = cancer,
```

```r
      method = "glm",
      family = "binomial",
      metric = "ROC",
      trControl = control
    )

    print(cv_model)
```

```
## Generalized Linear Model
##
## 568 samples
##  30 predictor
##   2 classes: 'Benign', 'Malignant'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 455, 454, 454, 454, 455
## Resampling results:
##
##   ROC        Sens       Spec
##   0.9640017  0.9494523  0.9431894
```

While the model performed well, the presence of highly correlated predictors suggested redundancy in the feature set. This motivated the use of regularization to improve interpretability and reduce the number of predictors.

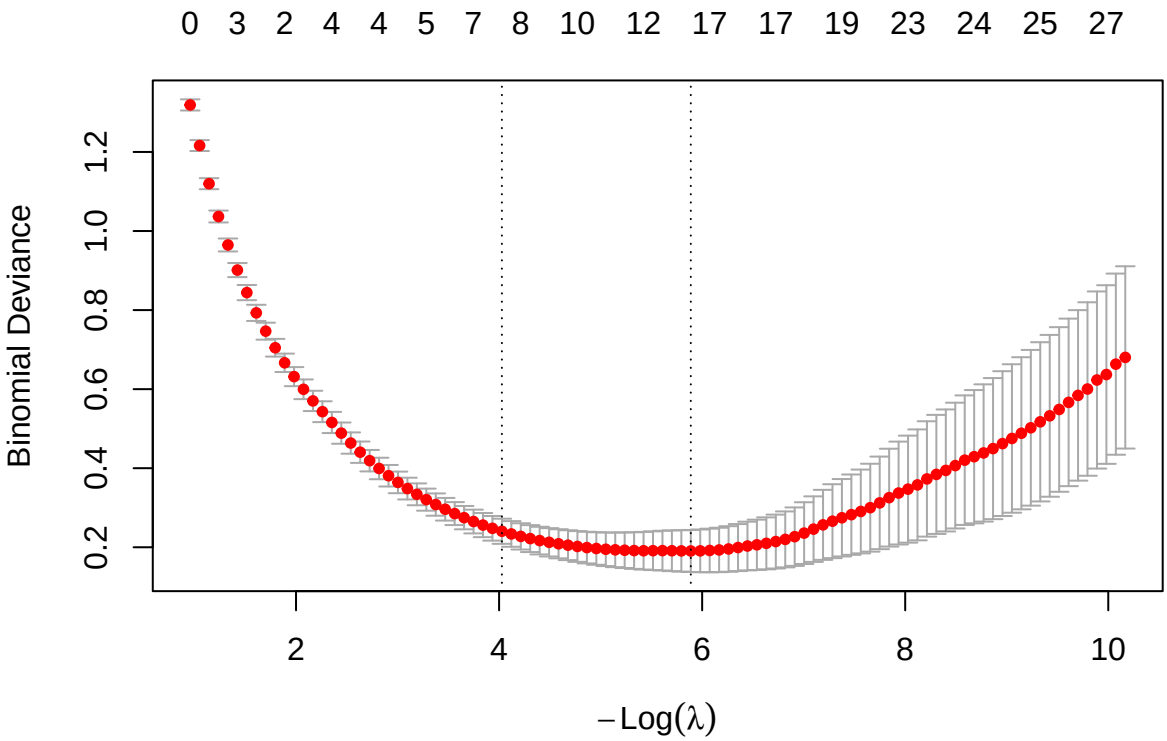
## Regularized Logistic Regression

```r
x <- model.matrix(diagnosis ~ ., cancer)[, -1]
y <- cancer$diagnosis

set.seed(123)

cv_glmnet <- glmnet::cv.glmnet(
  x = x,
  y = y,
  family = "binomial",
  alpha = 1
)

plot(cv_glmnet)
```




```r
print(cv_glmnet$lambda.min)
```

```
## [1] 0.002772109
```

```r
print(cv_glmnet$lambda.1se)
```

```
## [1] 0.01781932
```

To reduce redundancy among predictors and improve model interpretability, a regularized logistic regression model using Lasso (L1) regularization was implemented.

Lasso regularization adds a penalty to the model that shrinks less important coefficients toward zero, effectively performing feature selection. This is particularly useful in this dataset, where many predictors are highly correlated.

The model was trained using cross-validation to identify the optimal level of regularization. The results showed that a range of regularization strengths produced similar performance, indicating that a simpler model could be used without sacrificing predictive accuracy.

Using the more regularized model (lambda.1se), the number of predictors was reduced substantially, while maintaining strong classification performance. This confirms that many of the original features were redundant and that a smaller subset of variables is sufficient for accurate prediction.

The features retained in the regularized model were primarily related to tumor shape and boundary irregularity, such as concave points, concavity, and symmetry. This suggests that morphological complexity plays a more important role in distinguishing malignant tumors than size alone.

```r
coef_matrix <- coef(cv_glmnet, s = "lambda.1se")

nonzero_coef <- coef_matrix[coef_matrix[,1] != 0, , drop = FALSE]

cat("=== Selected Features (Lasso - lambda.1se) ===\n")
```

```
## === Selected Features (Lasso - lambda.1se) ===
```

```r
print(nonzero_coef)
```

```
## 9 x 1 sparse Matrix of class "dgCMatrix"
##                        lambda.1se
## (Intercept)          -16.1356411
## concave_points_mean   13.4176041
## radius_se              1.1199620
## radius_worst           0.4633270
## texture_worst          0.1190621
## smoothness_worst       7.8337451
## concavity_worst        0.1042767
## concave_points_worst  16.7278833
## symmetry_worst         2.5946527
```

## Final Model Evaluation

```r
x_test <- model.matrix(diagnosis ~ ., test)[, -1]


probs_glmnet <- predict(cv_glmnet, newx = x_test, s = "lambda.1se",
type = "response")


preds_glmnet <- ifelse(probs_glmnet > 0.5, "Malignant", "Benign")
preds_glmnet <- factor(preds_glmnet, levels = levels(test$diagnosis))


caret::confusionMatrix(preds_glmnet, test$diagnosis, positive = "Malignant")
```

```
## Confusion Matrix and Statistics
##
##            Reference
## Prediction  Benign Malignant
##   Benign        70         1
##   Malignant      1        41
##
##                Accuracy : 0.9823
##                  95% CI : (0.9375, 0.9978)
##     No Information Rate : 0.6283
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.9621
##
##  Mcnemar's Test P-Value : 1
##
##             Sensitivity : 0.9762
##             Specificity : 0.9859
##          Pos Pred Value : 0.9762
##          Neg Pred Value : 0.9859
```

```
##              Prevalence : 0.3717
##          Detection Rate : 0.3628
##    Detection Prevalence : 0.3717
##       Balanced Accuracy : 0.9811
##
##        'Positive' Class : Malignant
##
```

The regularized logistic regression model was evaluated on the test set to assess its performance on unseen data.

The model achieved high classification accuracy, with strong sensitivity and specificity. Only a small number of observations were misclassified, indicating that the model performs reliably across both classes.

In addition to accuracy metrics, the model demonstrated near-perfect discrimination ability, with an area under the ROC curve (AUC) of 0.9993. This confirms that the model is highly effective at distinguishing between benign and malignant tumors.

Importantly, this level of performance was achieved using a reduced set of predictors, demonstrating that a simpler model can perform just as well as a more complex one.

```r
probs_glmnet <- as.numeric(probs_glmnet)
roc_obj_glmnet <- pROC::roc(test$diagnosis, probs_glmnet)
```

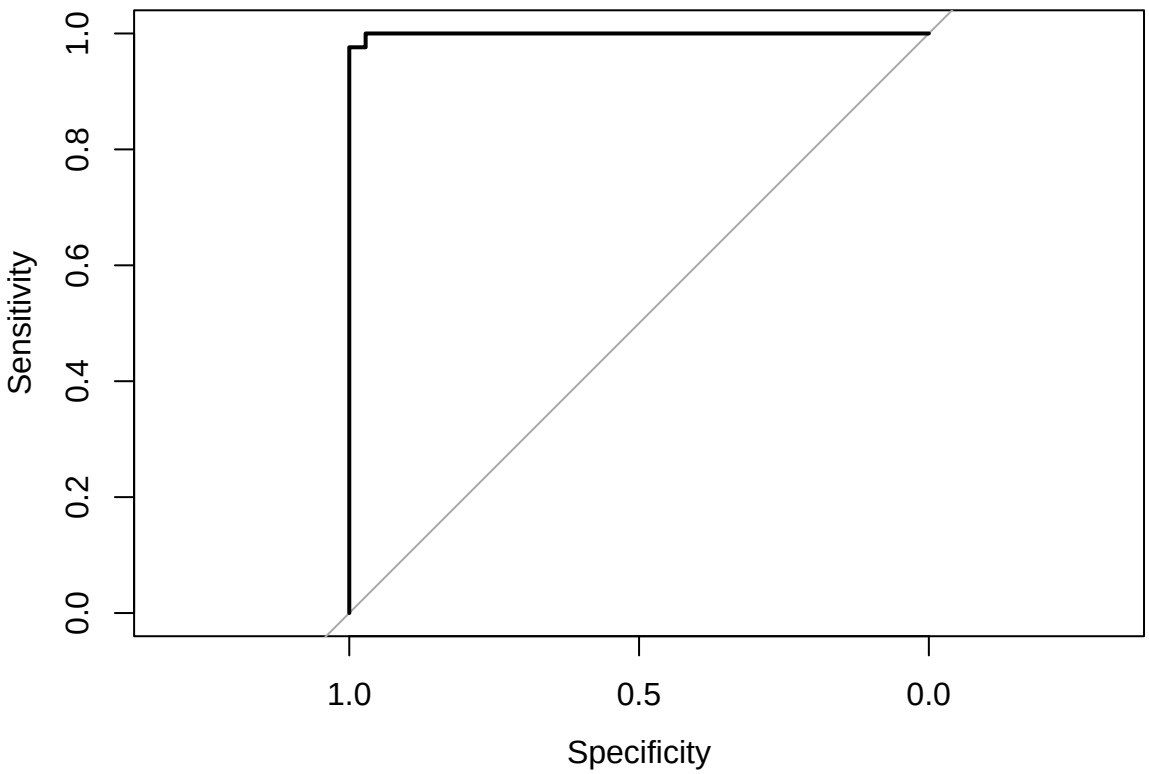
```
## Setting levels: control = Benign, case = Malignant
```

```
## Setting direction: controls < cases
```

```r
pROC::auc(roc_obj_glmnet)
```

```
## Area under the curve: 0.9993
```

```r
plot(roc_obj_glmnet)
```




##

Conclusion

This project demonstrated that breast cancer tumors can be accurately classified using a relatively small subset of features derived from tumor measurements. It is important to note that this dataset exhibits strong separation between benign and malignant tumors, which contributes to the high performance observed across models. As a result, the near-perfect classification metrics may not fully reflect performance on more complex or less separable real-world data.

While a standard logistic regression model achieved strong performance, the presence of highly correlated predictors highlighted redundancy in the feature set. Cross-validation confirmed that the predictive signal in the dataset was robust, while the correlated predictors reinforced the value of a regularized approach for feature selection and interpretability.

By applying Lasso regularization, the model was simplified significantly, reducing the number of predictors while maintaining high classification performance. The final model achieved strong accuracy, sensitivity, and specificity, along with near-perfect discrimination as measured by AUC.

Importantly, the results showed that features related to tumor shape and boundary irregularity played a more significant role in classification than size alone. This highlights the value of feature selection and regularization in building interpretable and reliable predictive models.